PRISM (Privacy-Restricted Information for Subscriber Modeling)

SENG474/CSC503 Project Proposal

Topic: Evaluating Privacy-Constrained Feature Subsets for Customer Churn Classification

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1. Problem Formulation

Customer churn prediction is formulated as a **Boolean Classification** task under a **Fully Supervised Learning** approach. Let O denote the set of customer observations, and let $C = \{0, 1\}$ be the finite set of classes:

- $c = 1$ (Positive Class): The customer churns (cancels service).
- $c = 0$ (Negative Class): The customer stays (retains service).

Our goal is to learn a mapping function $f : O \to C$ such that $f(o) = c$ for each customer observation $o \in O$.

2. Proposed Solution & Privacy Tiers

Standard churn models rely on granular behavioral tracking, such as daily app usage and interaction logs. However, modern privacy regulations and cold-start scenarios often restrict access to this activity data. In this project, we evaluate how accurately models can predict churn when access to user features is intentionally restricted. We stratify the dataset into three privacy tiers:

1. **Tier 1 (Private / Zero-Telemetry):** Basic account and billing metadata only (tenure, contract type, payment method, paperless billing).
2. **Tier 2 (Service Configuration):** Subscribed features (internet type, tech support, streaming add-ons) without usage frequency logs.
3. **Tier 3 (Full Profile):** Complete historical logs, total charges, and customer demographic attributes.

We aim to measure the performance degradation curve across these tiers to determine if basic billing metadata is sufficient to identify at-risk subscribers.

3. Dataset & Baseline Techniques

- **Dataset:** We use the public **Telco Customer Churn Dataset** (7,043 records, 21 attributes). Continuous features will be standardized, and categorical variables will be one-hot encoded.
- **Baselines:** We benchmark four distinct classification techniques across each tier: **Logistic Regression** (linear), **K-Nearest Neighbors (KNN)** (distance-based), **Random Forest** (ensemble tree), and a **Multi-Layer Perceptron (MLP)** (neural network). We will also test **Principal Component Analysis (PCA)** on Tier 3 features to see if low-dimensional projections preserve predictive signal.

4. Experimental Evaluation Plan

Because churners make up a minority of the dataset ($\sim$26%), raw accuracy is uninformative. Models will be evaluated using stratified 5-fold cross-validation. To handle class imbalance during training, we will use cost-sensitive class weighting. Our primary evaluation metrics will be **Precision-Recall AUC (PR-AUC)** and the **F_2-Score** (prioritizing recall to catch churning subscribers over precision).

5. Repository & Reproducibility

All code, modular directory structures, experiment logs, and a 20-row toy dataset for rapid testing will be maintained at: https://github.com/ibac03/PRISM-churn-privacy.